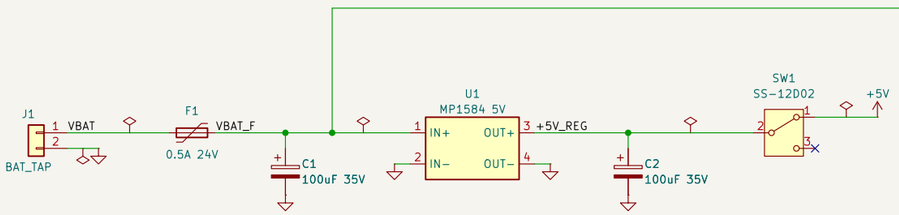
POWER (3S 12.6V to 5V). Motor current does not pass through this PCB

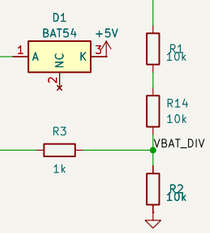


OPEN: J1 connector type; SW1, U1 footprints (measure first); D1 pinout (check purchased part datasheet)

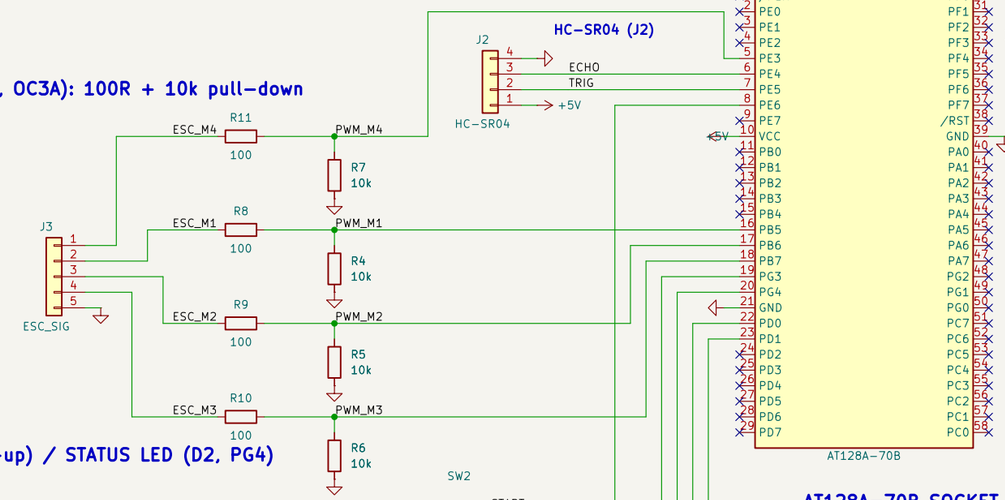
VBAT SENSE -> PF0

10k+10k / 10k: 12.6V -> 4.2V

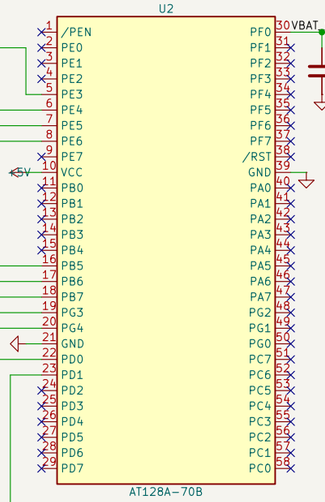
1k + BAT54 clamp + 100nF



ESC SIGNAL (OC1A/B/C, OC3A): 100R + 10k pull-down

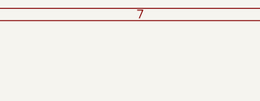


HC-SR04 (J2)

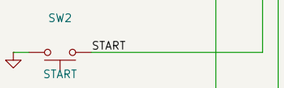


DECOUPLING 100nF

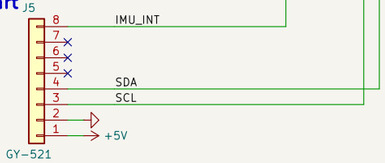
C3 U2, C5 J2, C6 J5 — each placed at its part on PCB



START BUTTON (SW2, PG3, internal pull-up) / STATUS LED (D2, PG4)



GY-521 (J5): VCC GND SCL SDA XDA XCL ADO INT — verify pin order on part



AT128A-70B SOCKET. Not used: PF4-7 (JTAG), PE0/PE1/PB1 (ISP), USART0

Sheet: /		
File: drone_v2.kicad_sch		
Title: ATmega128 Quadcopter V2 — Autonomous Carrier PCB		
Size: A3	Date: 2026-11-10	Rev: V2 draft
KiCad E.D.A. 10.0.6		Id: 1/1